

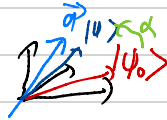
Arbitrary state preparation

Input: $|\psi_0\rangle = |0\rangle$

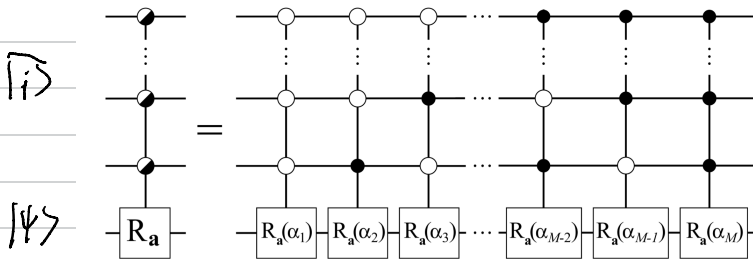
Output: $|\psi\rangle = [a; e^{i\omega_i}]^T$

① Single-qubit rotation $R_{\vec{a}}(\alpha)$

$$R_{\vec{a}}(\alpha) = e^{i\vec{a} \cdot \vec{\sigma} \alpha/2} = \cos \frac{\alpha}{2} + i(\vec{a} \cdot \vec{\sigma}) \sin \frac{\alpha}{2}$$



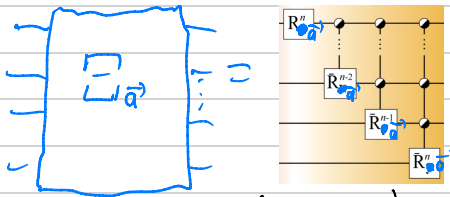
② k-controlled, m-target bit rotation $F_m^k(\vec{a}, \vec{a})$



$$F_m^k(\vec{a}, \vec{a}) \left(\left(\sum_i a_i |i\rangle \right) \otimes |\psi_0\rangle \right) = \sum_i a_i (|i\rangle \otimes R_{\vec{a}}(\alpha_i) |\psi_0\rangle)$$

$$F_m^0 = R_{\vec{a}}(\alpha_0)$$

③ Multi-qubit rotation $\Xi_{\vec{a}}(\{\vec{a}_i\})$



$$\Xi_{\vec{a}}(\{\vec{a}_i\}) = \prod_{i=1}^n F_i^{j-1}(\vec{a}, \vec{a}_{n-i+1}) \otimes I_{2^{n-i}}$$

④ Arbitrary state preparation

$$\textcircled{c} (a_i, w_i) = \sum_z \{ \vec{a}_i^z \} \sum_y \{ \vec{a}_i^y \} \sum_{a_i, l_i} \sum_{a_i, e^{i w_i} (i)} |\bar{0}\rangle - \sum_y - \sum_z - |\psi\rangle$$

Thus, we need to figure out $\vec{z}_i^y(q_i)$, $\vec{z}_i^z(w_i) = ?$

By solving the recursion, (which I have no idea how to solve)

$$\alpha_{j,k}^y = 2\sin\left(\sqrt{\sum_{l=1}^{2^{k-1}} |a_{(2j-1)2^{k-1}+l}|^2} / \sqrt{\sum_{l=1}^{2^k} |a_{(j-1)2^k+l}|^2}\right),$$

where $j = 1, 2, \dots, 2^{n-k}$ and $k = 1, 2, \dots, n$. Fig. 3 shows the quantum circuit corre

$$\alpha_{j,k}^z = \sum_{l=1}^{2^{k-1}} (\omega_{(2j-1)2^{k-1}+l} - \omega_{(2j-2)2^{k-1}+l}) / 2^{k-1},$$

where $j = 1, 2, \dots, 2^{n-k}$ and $k = 1, 2, \dots, n$.